AquaLens: AI-Powered Transparency for the Informal Water Market

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Problem Statement

The informal urban water tanker market operates as an opaque, unregulated cartel. Consumers suffer from profound information asymmetry, leaving them highly vulnerable to predatory surge pricing and the covert delivery of chemically hazardous, hyper-calcified groundwater that silently deteriorates residential infrastructure.

Problem Understanding

- **Vulnerable Stakeholders:** Resident Welfare Associations (RWAs), municipal facility managers, and university hostels bearing the financial brunt of volatile water acquisition.
- **Systemic Root Causes:** Rapid aquifer depletion forces private vendors to draw from deeper, unmapped geological fractures, causing extreme day-to-day fluctuations in Total Dissolved Solids (TDS) and pH levels.
- **Socio-Economic Consequences:** A total absence of verifiable supply-chain data forces consumers to rely on "blind trust." This enables price gouging during seasonal droughts (often 300% markups) and causes chronic dermatological and infrastructural harm.

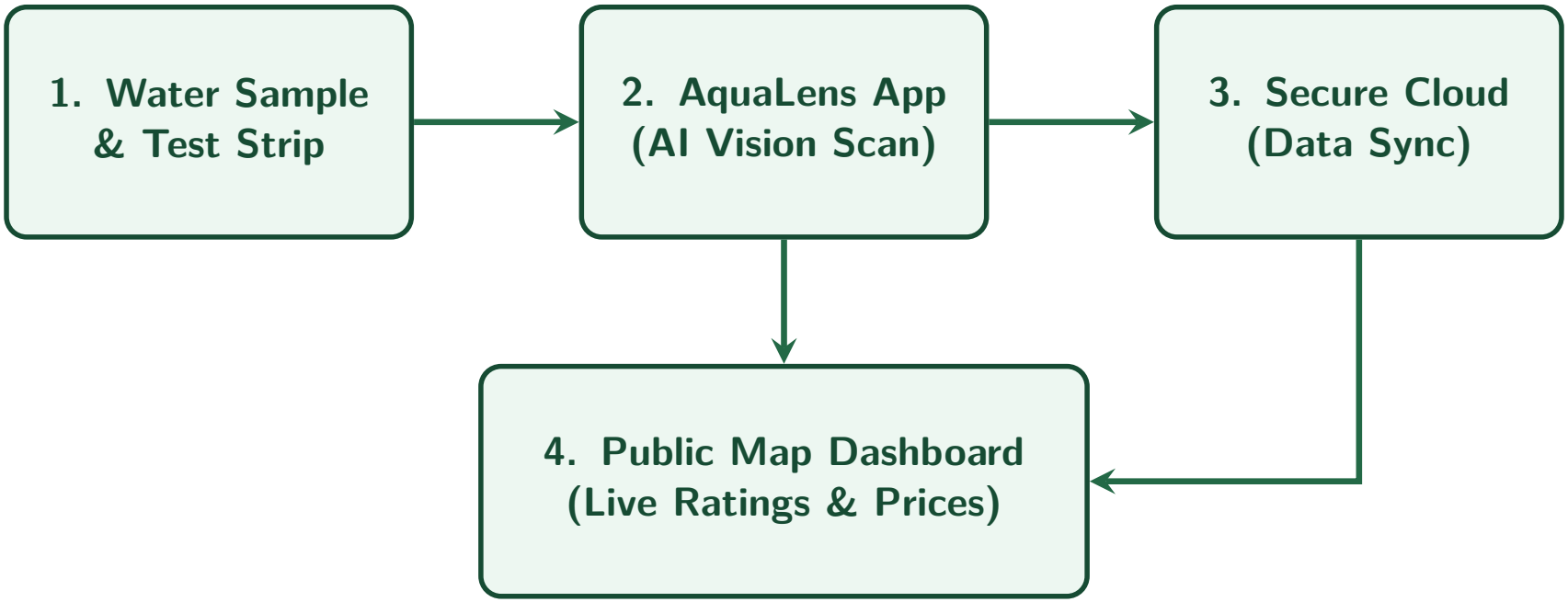
Existing Gaps & Challenges

- **Logistics Platforms Lack Quality Assurance:** Existing apps (e.g., TankerTap) act purely as ride-sharing for water trucks; they offer zero point-of-sale chemical validation, solving only delivery, not safety.
- **Prohibitive CapEx Requirements:** Traditional IoT water monitoring relies on expensive hardware sensors (*Rs.*15,000+) or high-latency lab tests, making real-time, decentralized monitoring economically impossible.
- **Absence of a Public Ledger:** There is no centralized economic framework to track historical vendor compliance, preventing ethical vendors from building reputation and allowing bad actors to exploit localized scarcities.

Our Proposed Solution

- **The Innovation:** *AquaLens* — A Zero-CapEx smartphone colorimeter and decentralized spatiotemporal ledger that democratizes water quality verification.
- **Scientific Execution:** Users dip an ultra-cheap (*Rs.*10) reagent strip into the delivered water. The AquaLens app uses Edge-AI computer vision to normalize ambient lux variations, mapping the strip's RGB chromaticity to precise pH and Hardness indexes.
- **Socio-Economic Impact:** Instantly empowers consumers to mathematically reject hazardous water at the gate, while crowdsourcing the data into a public "Yelp-style" heat map to permanently stabilize regional pricing.

Technical Approach



OpenCV normalizes ambient lighting to accurately extract HSV/RGB values, ensuring scientific consistency without external hardware.

Tech Stack

Frontend: React Native / Flutter

Backend: Python (FastAPI)

Database: Firebase Firestore (Real-time)

AI/ML: OpenCV (Computer Vision Colorimetry)

Innovation & Uniqueness

- **Translating Academic Research:** We are bringing cutting-edge 2026 academic smartphone colorimetry out of the sterile laboratory, engineering it into a hyper-local, fault-tolerant consumer application.
- **Zero Custom Hardware (Zero-CapEx):** By relying purely on the community's existing smartphone CMOS sensors and off-the-shelf aquarium test strips, we achieve extreme scalability that hardware-heavy startups cannot match.
- **First-to-Market Socio-Economic Integration:** AquaLens is the first solution to bridge the gap between hard chemical science and urban economics, fusing water quality validation with a live municipal pricing ledger.

Feasibility & Scalability

- **Feasibility (High):** The underlying OpenCV color extraction logic is mathematically proven. Our execution requires no hardware manufacturing—only software engineering and localized algorithm training.
- **Scalability (Frictionless):** The cloud-native architecture allows AquaLens to scale immediately from Coimbatore to Chennai or Bengaluru. Adding a new city requires zero physical infrastructure deployment.
- **Core Risk & Algorithmic Mitigation:** *Risk:* Variable ambient lux levels skewing camera color perception. *Mitigation:* The UI enforces a fixed-reference white-balance lock via the smartphone flashlight and utilizes an on-screen focal reticle.

Cost for Product

- **Operational Cost for End-Users:** *Rs.*0 for application access. The only recurring variable cost is the physical 5-in-1 chemical test strips, bulk-sourced at approximately *Rs.*8 to *Rs.*10 per test.
- **System Hosting & Maintenance:** Firebase Free Tier covers the entire R&D and pilot phase. Scaling to a municipal level requires an estimated \$20-\$40/month via GCP/AWS.
- **MVP Development Budget:** Less than *Rs.*2,500 total, allocated entirely toward procuring chemical test strips from commercial suppliers for rigorous algorithm calibration and field trials.

Use Cases (compulsorily 3 use cases must be there)

- **Use Case 1: The RWA Facility Manager (Infrastructural Defense)**
  A gate watchman scans an arriving 12,000L tanker. The AI detects highly calcified water (Hardness $>$ 500ppm). The delivery is instantly rejected, preventing severe scaling in the building's plumbing network.

- **Use Case 2: The Drought-Stricken Resident (Combating Surge Pricing)**
  During peak summer scarcity, a resident utilizes the AquaLens heat map to identify vendors offering standard pricing (*Rs.*800), bypassing the predatory surge prices (*Rs.*2,500) enforced by unverified local cartels.

- **Use Case 3: The Ethical Vendor (Market Incentivization)**
  A high-quality supplier encourages customers to scan their deliveries. Consistently optimal readings earn the driver a "Verified Premium" digital badge, dramatically increasing their B2B contracts.

Impacts and Benefits

- **Micro-Economic & Health Defense:** Eradicates the massive financial burden of replacing corroded residential plumbing while protecting consumers from dermatological and gastrointestinal issues linked to untreated groundwater.
- **Macro-Socioeconomic Market Correction:** Dismantles predatory tanker cartels by introducing a transparent, community-driven economic ledger. This forces open-market competition that permanently stabilizes surge pricing during droughts.
- **Expected Measurable Outcomes:** A projected 30-40% reduction in RWA water acquisition costs during summer months, alongside a quantifiable baseline of pH/TDS compliance across all mapped municipal deliveries.

Sustainable Development Goals (SDGs)

SDG	Target	Goal Name	Scientific & Economic Connection
SDG 6	6.1	Clean Water & Sanitation	Democratizes quality metrics via Edge-AI, ensuring equitable access and breaking vendor monopolies.
SDG 11	11.6	Sustainable Cities	Digitizes informal tanker logistics into a live ledger, enhancing urban resilience against water scarcity.
SDG 3	3.9	Good Health & Well-being	Mitigates exposure to hazardous pathogens and calcification, acting as a decentralized public health defense.

Individual Member Contribution

Member	Contribution or responsibility
Abishek N	Project Lead (Research & Idea suggestion)
Anamika V M	App design and came up with the camera clipper idea
Ananya L	Created the project flowcharts and planned the core mechanism
Eshaan Parihariya	Improved the overall concept and brainstormed UI ideas

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